

1	(a)	(i)	As the wrench falls, it loses gravitational potential energy but it gains kinetic energy. Kinetic energy = $\frac{1}{2}$ mass $\times$ square of speed, therefore its speed increases. Centripetal acceleration is directly proportional to the square of the velocity and inversely proportional to the radius of the circular path, which in this case is the radius of the dome, and is constant. Therefore, the centripetal acceleration of the wrench increases as it slides from A to B.	
		(ii)	N = 0	
	(b)		Since N = 0 when wrench loses contact with the surface, $mg \cos \theta = \frac{mv^2}{r}$ $mg\left(\frac{h}{R}\right) = \frac{mv^2}{R}$ $h = \frac{v^2}{g}$	
	(c)	(i)	$v^2 = hg$ $v^2 = \frac{2}{3}(30.0)(9.81)$ $v = 14.0 \text{ m s}^{-1}$ We also have, $\cos \theta = \frac{h}{R}$ $\cos \theta = \frac{\frac{2}{3}R}{R} = \frac{2}{3}$ $\theta = 48.2^\circ$	
		(ii)	Considering vertical components and taking vectors downwards as positive, $s_y = u_y t + \frac{1}{2} g t^2$ $h = \frac{2}{3} R = (v \sin \theta) t + \frac{1}{2} (9.81) t^2$ $h = \frac{2}{3} R = (14.0 \sin (48.2)) t + \frac{1}{2} (9.81) t^2$ $\frac{2}{3} (30.0) = (10.44) t + \frac{1}{2} (9.81) t^2$ $20.0 = (10.44) t + \frac{1}{2} (9.81) t^2$	

			$t = 1.22 \text{ s}$	
		(iii)	$s_x = u_x t + \frac{1}{2} a_x t^2$ $s_x = v \cos \theta(t)$ $s_x = 14 \left(\frac{2}{3} \right) (1.22)$ $x = 11.4 \text{ m}$	

2	(a)	(i)	$W = p \Delta V = 57.0 \times 10^5 \text{ Pa} \times (2.1 - 2.0) \times 10^{-5} \text{ m}^3$	
---	-----	-----	--	--